Reality and Consciousness Are Mathematical Structure

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Abstract

Everything that exists can only be mathematical structure, because whatever exists has definite features, every definite feature is either ruled or random, and both are mathematical. The view is monism: one kind of thing throughout. It is not Platonism, which sets the Forms apart from a world they do nothing in, a dualism with a supernatural second realm. Plato's One, with no size or properties, is nothing at all, and Neoplatonic emanations only multiply the problem of how anything could relate to it. The hard problem of consciousness is a fallacy that assumes its conclusion, that sentience cannot be structure, and then finds that structure cannot explain it. Qualia, panpsychism and idealism each need the structure they set out to supplement. The strongest objection says structure leaves out an intrinsic nature. A swap test answers it. Exchange the hidden natures behind mass and charge while every relation stays fixed, and nothing anyone could notice, think or say would change. Such a nature could not even be known by the mind it supposedly makes conscious. The claim is an a priori proof in the logician's sense, necessary as 2 + 2 = 4 is, and it has teeth: one coherent feature neither ruled nor random would refute it. Consequences follow by reasoning alone. Every consistent structure exists, fictions included, with every plot hole fixed by some consistent math. Each added bit halves a description's weight, so simple laws dominate. Complex minds must grow from a simple start in causal steps, and time is those steps, running forward.

Keywords: mathematical monism, mathematical structure, structural realism, Platonism, Neoplatonism, hard problem of consciousness, panpsychism, idealism, Level IV multiverse

1. Introduction

Everything that exists can only be mathematical structure, because whatever exists has definite features, every definite feature is either ruled or random, and both are mathematical. That is the core claim of Matheism and the claim defended here. Math means the structures themselves: their properties, relations and possible changes, simple or complex, predictable or random. Equations are ways we describe them. A triangle is a geometric structure, not something made by geometry. Reality stands in the same relation to mathematics.

The view is monism, the view that reality is one kind of thing: *Deus sive Mathematica*, God or mathematics. It is not Platonism, which is dualism. Plato set the Forms, the perfect patterns of things, apart from the world, where by his own student's account they cause no change at all (Aristotle, *Metaphysics* I.9). A realm outside nature that does nothing in nature is supernatural in the plain sense of the word. Plato's highest principle fares worse. A One so pure that it has no size or properties is nothing at all, and the emanations Plotinus added only multiply the problem, since each level must relate to a One that has nothing to relate by. Matheism takes the side of the Pythagoreans, for whom things are numbers (Section 3).

The hard problem of consciousness asks why physical processes come with felt experience (Chalmers, 1996). As posed, it is a fallacy that assumes its conclusion: it takes sentience to be something structure cannot be, then finds that no structure explains it. It expects to feel a structure from outside, by reading about it; a structure is felt only by being it. Qualia, the felt qualities of experience taken as something over and above structure, only label the gap. Panpsychism,

the view that all matter has some form of mind, cannot say how tiny feelers add up to one mind without a law of combination, and a law is structure. Idealism, the view that reality is made of mind, cannot say how separate minds discover the same mathematics without using mathematics. They need math, and math does not need them (Section 4).

The hardest objection grants the mathematical account and says each thing still has an intrinsic nature, what it is in itself apart from its relations, which the account merely represents. Answering it is the main work of the paper. The answer is a swap test. Exchange two such natures while every relation stays fixed, and ask whether anything anyone could notice, think or report has changed. If something changes, the nature does work and belongs to the structure. If nothing changes, nobody could know the nature, including the people it supposedly makes conscious (Sections 2.2 and 4.3).

The claim is necessary, as 2 + 2 = 4 is, and it still has teeth. It rules out every rival that needs something with no structure, and those rivals have serious defenders: a bare substrate, a soul that moves a body by no law, a free choice neither caused nor random, a creator outside mathematics. One coherent feature that is neither ruled nor random would refute it.

Consequences follow by reasoning alone, with no physics beyond what everyone grants. Every consistent structure exists, fictions included, and every plot hole is fixed by some consistent math. An unruled bit is a 50/50 choice, so each added bit halves a description's weight, and simple laws dominate: ours fit on a page. Minds like ours are complex, so they must grow from a simple start, and growth takes causal steps, because a cause needs a next state and a continuum has none, an a priori proof in the logician's sense. Time is those steps, so it runs forward. Quantum theory does not force uncaused chance on us: on the many-worlds and superdeterministic readings, every outcome follows a rule. The

identity thesis does not rest on these consequences, and each is stated below with its premise.

Ontic structural realism, the view that structure itself is the basic reality, is the contemporary starting point (Ladyman and Ross, 2007; French, 2014, 2019). Stathis Psillos (2001) and Anjan Chakravartty (2007) object that structure needs an underlying bearer. Baruch Spinoza's monism and necessity offer a comparison, not this exact identity (*Ethics* I.14, I.29, I.33, I.34). The nearest physical proposal is Max Tegmark's (2008) Mathematical Universe Hypothesis, that physical reality is a mathematical structure, which he derives from the existence of an external reality independent of human observers. The argument here starts instead from what any definite nature must be.

The contribution is that identity argument: whatever exists is some definite way, every definite way is ruled or random, and an intrinsic nature that makes a determinate difference belongs to the structure, while one that makes none constitutes nothing. The decisive premise is that a complete constitutive structure leaves a substrate nothing to contribute. It rules out a bare substrate, intrinsic natures that do no work, a mind with a nonstructural core, and a second realm, whether of Forms or of a One beyond all properties.

2. Why Nothing Can Fall Outside Math

2.1 What Would a Nonmathematical Thing Be?

For anything to exist and be distinguishable from nothing, it must have determinate features. 'Something' differs from 'nothing' by having properties: extent, duration, charge, mass, complexity. These cannot be wholly arbitrary or undefined. Existence requires determinacy; determinacy requires distinguishing features; distinguishing features require relations of sameness and difference.

Properties do not float free. Each is fixed by how it relates to others. A position is where something lies relative to other positions. A mass is how hard a thing is to push and how strongly it pulls on other masses. A price is what it trades for. Take away every relation a property enters, and nothing is left to say what it was. Physics has made structural description ever more central (Ladyman and Ross, 2007; Esfeld and Lam, 2008), and the argument needs only the general point: to be a definite feature is to stand in definite relations.

2.2 Why the Complete Structure Is the Thing

Showing that reality is mathematical structure means showing that nothing could distinguish the two, and the argument starts from sameness of form. Any relational structure is isomorphic to at least one mathematical object, where an isomorphism is a structure-preserving mapping: it preserves all relations while potentially changing labels. Many mathematical objects share any given form, and the identity claimed here is with that shared form, whose relata, the things it relates, are simply its places.

What distinguishes a physical relational structure from an isomorphic mathematical structure? Not their relationships, identical by definition of isomorphism. Not their causal powers: causal relations hold between parts of a structure, and an isomorphism preserves them, so only the whole lacks causes outside itself, as the physical universe does. Not their observable properties, which reduce to relational structure. The only remaining candidate is non-relational 'substrate,' some primitive (taken as basic, with no further analysis) 'physical stuff' that bears properties without being constituted by them.

The strongest opponent grants every mathematical relation and still assigns the things related an intrinsic nature. Giving that nature a symbol does not settle whether it is mathematical. The question is what the symbol stands for. A mass, a

power or a felt quality is a proposed constituent with a character of its own; it cannot be dismissed simply because it is unobservable. Matheism argues that any such character belongs to the complete structure of the thing. A supposed bearer left over after every constitutive feature has been included supplies no further nature to bear.

This is the claim that structure is complete, sometimes called structural exhaustion. It is stronger than saying an extra ingredient is unnecessary for prediction. The opponent may still insist that a property has primitive nonstructural content. Matheism's challenge is to give that content without either specifying more structure or merely asserting a remainder. The argument turns on that challenge, not on the existence of a mathematical encoding.

The proposed substrate faces three objections. First, a complete account of a thing's relations leaves a bearer no job. Physical theory specifies states, fields and interactions without a separate variable for featureless stuff that bears them. Reading that account as the complete constitution of reality is the structuralist interpretation defended here. The claim concerns a complete organized structure, not the replacement of a thing by an isolated number.

Second, substrate does no explanatory work. Two systems with identical relational structure behave identically in all observable ways. The substrate makes no difference to anything we can measure, predict or explain, so positing it violates Occam's Razor. Any proposed substrate must itself have determinate features, or it is nothing, and determinate features constitute structure. By the rule-random dichotomy, substrate *is* structure, the very thing it was meant to underlie. This applies at every level: no level of decomposition yields a nonmathematical residue.

Third, a primitive 'this one, not that one' is substrate under another description. Haecceitists invoke haecceities, primitive thisnesses, to distinguish entities when

all relational properties match. The haecceitist must specify what makes *this* haecceity differ from *that* one. Any answer appeals to distinguishing features, which are properties. Declining to answer leaves the distinction primitive and unexplanatory: a label where an explanation should go.

A complete description must preserve every constitutive relation, including relations to an environment and to observers. Even two internally isomorphic systems can be numerically distinct within a larger structure. In the universe of two identical spheres that Black (1952) describes, each sphere stands at a distance from the other and at none from itself, so a relation shows that there are two. The identity argument asks about the complete reality: what, if anything, could distinguish it from its complete structure? An additional relation enlarges the structure. A claimed difference that carries no determinate feature adds nothing to what constitutes the thing. The math is the territory.

The complete relational structure exhausts the properties and causal roles of a physical reality, so no further substrate remains to constitute it. This is the identity Matheism defends: reality *is* mathematical structure. It does not instantiate, or embody, that structure in a separate nonstructural substance. Isomorphism specifies sameness of form; the completeness argument supplies the further ontological claim, one about what exists.

2.3 Ruled or Random: Why the Dichotomy Is Exhaustive

Every structure is of one of two types, ruled or random.

Ruled structures, or patterns, have low generative complexity: rules simpler than the structure itself generate them. The laws of physics are short equations that generate vast complexity. Conway's Game of Life generates endless patterns from four rules (Gardner, 1970). The digits of π form an endless sequence that does not repeat, generated by a finite formula. Fractals and deterministic chaos

belong here too. In each case a finite description generates or explains the structure, and apparent randomness can come from a compact rule: a finite stretch of π is generated by an algorithm and a request for that many digits.

The split between rule and randomness survives being made exact, and both sides stay mathematical. Algorithmic information theory measures the information in a string by the length of the shortest program that outputs it. Stated precisely, fix a universal description language U, a finite binary string s and a deficiency threshold c; then either $K_U(s) < |s| - c$ or $K_U(s) \geq |s| - c$. Both classes are mathematical, and the partition is exhaustive on that domain. No formal system, a set of axioms with rules for deriving theorems, can certify incompressibility beyond a fixed length (Chaitin, 1974), and the partition needs no certificate: every string falls on one side whether or not anyone can prove which. An incompressible string still has determinate entries and relations. Randomness for infinite sequences needs its own definition and can be noncomputable, and noncomputable objects are mathematical too.

Fundamental chance would leave an outcome undetermined by the prior state. A probability law can still specify the possibilities and their weights; it does not pick the history realized. If reason demands a sufficient reason for each outcome, brute selection is unacceptable, and on the view taken here, change without a cause lies outside logic and reason. The identity thesis does not need that premise: even granting chance, its law and its determinate outcomes are mathematical. Quantum theory, the usual home of claimed brute chance, does not force it on us. In the many-worlds interpretation every outcome occurs, and what looks like chance is which branch an observer finds itself in (Everett, 1957). In superdeterminism one rule correlates what is measured with how it is measured (Hossenfelder and Palmer, 2020). Bohmian mechanics, in which particles follow definite paths, is deterministic too (Bohm, 1952). None of them

adds a nonmathematical ingredient.

Four potential counterexamples deserve consideration. *Ontic vagueness*, vagueness in the world itself rather than in our language (Barnes, 2010): a theory may give a mathematical account of which facts are indeterminate without giving each vague property a sharp boundary. *Primitive dispositions*, powers such as fragility or solubility: manifestation conditions have structure, but a dispositionalist can dispute whether those conditions exhaust the power; this returns to the identity argument. *Non-algorithmic hypercomputation*, computing beyond what any computer program can do: noncomputable functions and oracle computations (which may consult answers that no program could compute) are mathematical objects too. Noncomputability is not nonmathematicality, and the arithmetical hierarchy, a ranking of number sets by the complexity of their definitions, does not exhaust all noncomputable objects. *Atomless 'gunky' structure*: still relational structure, just infinitely divisible. All coherent alternatives reduce to ruled or random mathematical forms.

The dichotomy is not a stipulation one is free to decline. Any objector must reason in a way that already sorts cases into the ruled and the random; an appeal to a domain "outside reason" is self-undermining, since the appeal itself uses the faculty it would exempt. Logic is not one commitment of inquiry among others; it is the medium in which all inquiry is conducted. This point has a narrow target: it answers the objector who denies that reality has mathematical structure at all. An objector who grants the structure and denies that it exhausts reality reasons mathematically without contradiction, and meeting him requires the identity argument of Section 2.2.

One might grant that reality is purely relational while denying it is thereby *mathematical*. Perhaps relations are primitive metaphysical facts that mathematics merely models? The response: rules express relations among

possible states. Probability theory specifies relations. A realized sequence has definite entries even when it admits no compression. 'Mathematical' does not mean regular, predictable or generated by a short program. Rule and randomness exhaust all possible relational character. 'Mathematical structure' is not a species of relation; it is a genus containing all relations. The genus is mathematics' own: in model theory, a branch of mathematical logic, a structure is any domain with relations defined on it (Hodges, 1993), so a determinate relational system counts as a mathematical structure before Matheism adds anything. The substantive step is the identity argument, that a complete structure is the thing itself (Section 2.2). The burden falls on the critic: produce a coherent third option beyond rule and randomness, or accept the conclusion.

2.4 Why Reliable Observation Requires Structure

A reliable record must track what it records, and a useful prediction must depend on evidence that bears on the event predicted. Observers who learn and act on records need these relations. The constraint concerns the relations that support observation, not the compressibility of everything observed. A memory can store an incompressible string with a simple recording rule, and a probability law can sustain stable structures and reliable statistics. It does not follow that every region of reality is compressible, that random data cannot be experienced, or that every apparent observer in a chance-driven model is impossible.

2.5 Newman's Objection and the Scope of Structural Claims

A bare claim that the world has some structure says too little, and the mathematician Max Newman showed why. His objection warns that a merely existential structural claim can become too weak. If relations may be freely reinterpreted, a domain of the right cardinality, or number of members, can be

equipped with relations realizing the stipulated abstract pattern (Newman, 1928). Merely adding more equations does not answer the objection if every term in those equations is equally open to reinterpretation.

The present proposal concerns the actual constitutive relations, including causal and modal organization: what interactions occur and how the system would respond under the specified variations. It does not infer a particular physical structure from cardinality alone. Empirical theory must identify and constrain those relations through observations. Whether that identification captures every intrinsic property remains the exhaustion dispute of Section 2.2.

This answers a different question from reconstructing knowledge through a Ramsey sentence, which replaces a theory's terms with variables. Newman's result trivializes the claim that some relations of a given form exist on a domain. The identity view makes a different claim: the world is one structure, its relations included, and observers, as parts of it, pick out those relations through their own relations to the rest. Which relations those are is the work of empirical theory. A theory's notation, coordinates and charts are structures too, and the identity concerns the structure its predictions depend on, not the notation chosen to state it.

2.6 Interaction Requires Shared Structure

Begin with a conversation. Suppose consciousness were an ineffable dimension beyond mathematics and beyond words. Then how would words reach it? Speech is sound waves. They change what a listener feels, and what the listener feels moves the listener's mouth and sends sound waves back: physics in, physics out. Somewhere the signal would have to leave the mathematics and translate back, and whoever places consciousness outside it owes an account of where and how.

For any two distinct substances or realms to interact, they must share common rules. If mind moves matter, or Forms act on particular things, some law must say how. In 1643 Princess Elisabeth of Bohemia asked René Descartes how a soul that only thinks could move a body (Shapiro, 2007). His pineal gland, a small gland deep in the brain, answered where, not how. Any answer has to say what changes, under which conditions and by how much. Those when-then relations are relational structure, and relational structure is mathematical. The step is a priori: to interact is to change by some rule, so no experiment is needed to show that interacting things share structure.

An interaction law places both proposed substances within one mathematical account. A dualist, who holds that mind is something beyond the physical, can still claim that the things related have an intrinsic nature the account does not exhaust. Matheism rejects that further claim through structural exhaustion: every constitutive difference must belong to the complete structure. An alleged extra substance faces two outcomes. If it contributes a further determinate feature, that feature extends the structure. If it contributes none, it plays no part in constituting the thing. Interaction makes the shared account unavoidable; the identity premise makes it sufficient.

3. Monism, Not Platonism

3.1 Plato's Second Realm

The objection that Matheism is sophisticated Platonism asks whether it posits two realms, abstract mathematics and concrete physical reality, with a mysterious bridge between them (cf. Benacerraf, 1973). It posits one. What we call physical is structure experienced from inside one part of it. What we call abstract mathematics is the same structure considered apart from any particular place in it.

The difference is one of perspective, not of kind.

Plato's own view has lasted through the ages, and so has belief in the supernatural; lasting makes neither true. The Pythagoreans held that things are numbers; Plato, Aristotle reports, held "that the Numbers exist apart from sensible things" (*Metaphysics* I.6, 987b27-29). He made mathematics the ladder away from the world: arithmetic, geometry, astronomy and harmony were to turn the soul from what it sees toward Forms no eye can see (*Republic* VII, 521c-531c). Matheism takes the Pythagorean side. Plato's Forms are not part of nature and do nothing in it. Asked what the Forms contribute to sensible things, Aristotle answered that "they cause neither movement nor any change in them" (*Metaphysics* I.9, 991a8-11). To say that things share in them, he added, "is to use empty words and poetical metaphors" (991a20-22). Plato needed a divine craftsman, the demiurge, to copy the Forms into the world (*Timaeus* 28a-29b).

A realm outside nature that does nothing in nature is supernatural in the plain sense of the word, and a view with two such realms is dualism. Plato's step also works against explanation itself, since what does nothing in the world explains nothing in it: it is a late Athenian anti-philosophy. Athens' early thinkers sought reason inside nature; Plato set the Forms outside it. Then something lies beyond reason. If it is the Forms, nothing here can know them. If it is the world, it is no Form and so no logic, a mystery box of chance and magic. If it is the link, no account can say how the two relate. In the end both halves turn supernatural: the Forms, because they stand outside nature, and nature, because without the Forms it follows no logic and cannot be known. Each option says that reality cannot be known, which is faith posing as reason, and an a priori proof shows it wrong: outside logic is illogical, as impossible as a square circle. A view that places reality beyond reason has exited philosophy: it refuses knowledge instead of loving it, and it cannot state itself without the logic it denies. An explanation

cannot be supernatural in any case, since to explain is to connect, and a connection is structure.

Plato saw that the Forms could not be known and wrote it down. In his *Parmenides*, the greatest difficulty for separate Forms is that we could not know them, and a god who has knowledge itself could not know our world (*Parmenides* 133b-134e). He kept the Forms anyway, since without them thought would have nothing to fix on (135b-c).

In Plato's own frame the split cannot stand. If the link between Forms and world could be understood exactly, it would follow a rule, and in Plato's terms what can be known is Form, so the link would be one more Form. A craftsman copying by such a rule would be applying Forms to Forms, and what he made would be Forms as well: one realm, which is Matheism. Plato did not go that far, though his *Timaeus* comes close: the craftsman builds fire, air, water and earth out of triangles (53c-56c). Where the geometry stops, Plato leaves a wandering cause that reason can only persuade (47e-48a) and a receptacle grasped by "a kind of bastard reasoning" (52b), a nature that follows no logic. Either the link is Form, and then the world is Form too, or it lies beyond reason, which is where Plato left it.

By Aristotle's account, Plato set the Forms apart to save knowledge from the flux. In his youth he learned from Cratylus, a follower of Heraclitus, that what is ever changing cannot be known, and he placed the objects of knowledge elsewhere (*Metaphysics* I.6, 987a32-b10). Heraclitus had already given the answer. His river is the same river while different waters flow through it (fragment B12): the pattern stays while the water moves. All things come to pass in accordance with the logos, the rule of the flow (fragment B1; Kahn, 1979). Plato took the flux without its logos and set the rule outside the world it rules.

Alfred North Whitehead called the European tradition a series of footnotes to Plato (Whitehead, 1978, p. 39), and whoever opposes Plato's dualism is said to remain in a dialectic with him. The opposition is a rejection of the rejection of pure reason. A case for pure reason reads as a reply to Plato only because he is its most influential early opponent. The divide is pure reason against supernatural nonsense, nonsense in the strict sense, since what lies outside logic cannot even be said. Descartes's two substances need a bridge no account can state, and Kant's thing in itself is reality declared unknowable (Kant, 1998, Bxxvi). Wherever the split appears, it puts reality beyond reason.

Contemporary Platonism keeps the split. An object counts as abstract "just in case it is non-spatiotemporal and (therefore) causally inefficacious" (Linnebo, 2023). Paul Benacerraf (1973) asked how creatures in space and time could know such objects. Section 2.6 asked how separate realms could act on each other, and Benacerraf asks the same of knowledge. Matheism has no second realm and so needs no bridge.

We know mathematical truths because we are mathematical structures, not because we have mysterious access to a separate domain. The question "where are mathematical structures located?" is a category error, treating one kind of thing as if it were another. Structures do not occupy places within a container; space and time are themselves structural relations. We exist in mathematical structure the way a triangle exists in geometry: as an instance of it, not as an object placed inside it.

3.2 A One Without Properties Is Nothing

Plato's highest principle fares worse than the Forms. The Good, he wrote, lies beyond being, surpassing it in rank and power (*Republic* 509b). The One of his *Parmenides*, taken strictly as one, has no parts, no shape, no place and no time.

Plato drew the consequence himself: such a One in no way is, and nothing can be said, known or perceived of it (*Parmenides* 137c-142a). A point so pure that it has no size or properties is nothing at all. It contains nothing, relates to nothing and affects nothing, so Plato's monism is fake: a dualism with an empty One on top. Section 2.1 reached the same verdict on any featureless bearer.

Plotinus, the third-century founder of Neoplatonism, made such a One the source of everything. It is neither a thing nor a quality nor a quantity, neither intellect nor soul (*Enneads* VI.9.3). From it emanates Intellect, from Intellect Soul, and from Soul the material world, each level flowing from the one above (V.1-2). The levels multiply the problem they were meant to solve: every level is more dualism, with the interaction problem again at each step down, and together they solve only the problems that dualism itself creates. Each must relate to the One, and a relation gives each of its terms a property, which the One cannot have. Calling the source One does not say how the levels connect. If they produce and act on one another, their relations are already structure, and the One above them adds nothing.

A Neoplatonist will answer that the One lies beyond being, not below it, as the power of all things (*dynamis pantōn*, III.8.10). A power is a relation to what it can produce, so the answer gives the One a property after all, and a One with that property is structure. Plotinus describes the procession itself in images. The One, seeking nothing and lacking nothing, overflows, and he adds *hoion*, 'as it were' (V.2.1); it gives off the lower levels as the sun gives off light and fire gives off heat (V.1.6). His own 'as it were' concedes the point: images are not relations. Aristotle's verdict on Plato's sharing applies to emanation as well, empty words and poetical metaphors.

The layers are unnecessary. Mathematical monism keeps one level, the structure, whose parts relate by the relations that make them what they are. Plotinus's One has no properties and so is nothing. The one that Matheism keeps has every

property there is, since it is all the structure there is.

4. Minds: The Hard Problem Assumes Its Conclusion

The strongest test of the identity thesis concerns minds. A rival account must say what its extra constituent contributes to experience or action. The challenge applies to substance dualism, mind as a separate nonphysical thing; to property dualism, physical things with nonphysical mental properties too; to panpsychism; to idealism; and to any intermediary between mind and matter.

4.1 The Fallacy

David Chalmers (1996) argues that even complete physical knowledge would leave unexplained why there is something it is like to be a conscious system, why structure comes with experience. This hard problem of consciousness motivates property dualism: something beyond structure must make up phenomenal consciousness, the felt side of experience.

As posed, the hard problem of consciousness is a fallacy that assumes its conclusion. Its premise is that experience is something other than structure. Granted that premise, no structural explanation can reach experience, and the problem looks hard. The premise is the very claim in dispute. A critic who stipulates that experience is nonstructural has disqualified every structural explanation in advance. The identity thesis rejects the stipulation, and asking whether structure and experience are identical does not assume that they are not. Seeing the circle takes no experiment, only the argument's own form, so the diagnosis is an a priori proof in the logician's sense.

The problem also expects the wrong kind of access. A structure is felt by being it, from inside. The hard problem of consciousness expects to feel it from outside, by reading a description, and that is a category error about what kind of thing

experience is. Any explanation, a dualist's included, would itself be more structure, so a gap drawn between structure and experience is closed to every theory alike. A problem that no possible answer could solve marks a stipulation, not a discovery.

The explanatory gap Joseph Levine (1983) identifies, that we cannot see why certain structures constitute experience, may reflect the structure of explanation rather than a gap in reality. In a dilute gas, temperature turned out to be the mean energy of motion of the molecules, measured on a different scale. Vitalism posited a nonphysical life force, *élan vital*; life turned out to be complex biochemistry. Water turned out to be H_2O. In each case the gap closed without a missing ingredient, when two descriptions turned out to pick out one thing.

The felt quality is what a conscious structure *is* from the inside: the character of being that particular pattern, not an additional property requiring separate explanation. This is an identity claim, not eliminativism, which denies that conscious experience is real. Being mathematical structure does not diminish the reality of experience; experience *is* the structure, known from within. Felt qualities are what mathematics feels like from inside.

4.2 Qualia, Property Dualism and Interaction

Qualia, taken as properties over and above structure, are a label for the gap, not a bridge across it. Property dualism adds them to physical properties, and the question is whether they affect what happens. If they have no physical effects, they cannot cause the report "I see red"; the physical process produces the report without them. If they do affect physical events, some law must relate the phenomenal and physical changes, and that law is structure (Section 2.6).

The retreat to epiphenomenalism, which treats consciousness as a causally inert byproduct, gives up the intuition that drove dualism, that our experiences cause

our reports of them. It also leaves our reports unexplained: if the feeling does nothing, our words about it track something else. Selection cannot favor an inert property for effects it does not have.

Software looks categorically distinct from hardware, yet it is constituted by hardware states; the difference lies in the description. The gap between a neural description and an experience is of the same kind. The experience is the organization itself.

4.3 Panpsychism, Russellian Monism and the Swap Test

Panpsychism cannot explain how tiny feelers add up to one mind. A combination principle would have to say which parts combine, under what conditions and with what resulting experience, and those conditions are structure (Chalmers, 2016).

Its most careful form, Russellian monism, defended by Galen Strawson and Philip Goff (Strawson, 2006; Goff, 2019), holds that physical science describes only relational structure, while the intrinsic nature underneath consists of experience. Bertrand Russell (1927) observed that physics describes only relational structure, leaving open what instantiates those relations. On mathematical monism this gap closes. Structure has no separate nature standing behind it, because structure is what there is. Strawson and Goff take consciousness seriously, and their error lies elsewhere, in positing something beyond structure to ground it.

A test case makes the dilemma concrete. Take the Russellian monist's own picture: mass is a role, whatever resists acceleration and attracts other masses, and the nature that fills the role is experiential. Now swap the natures behind mass and charge while every role stays the same. Every particle moves as before, every brain fires as before, and everyone says and believes exactly what they did before, including "my experience of red is like this."

If the swap changes anything, in what happens or in what anyone notices, thinks or reports, the nature does work in the organization and belongs to its structure. If it changes nothing, the people in both worlds hold the same beliefs about their experience, so those beliefs cannot be tracking the nature that differs (Lewis, 2009). A view that leaves everyone, its defenders included, unable to know the natures it posits loses its own motive, since knowing our experience is what led it to posit them. Lewis (2009) drew humility from the swap, accepting that we cannot know which nature fills which role, and kept the natures. The identity view takes the further step of Section 2.1. A determinate difference needs a distinguishing feature, and nothing marks which nature fills which role. So the two swapped worlds are one world, and the structure is all there is to the thing.

4.4 Idealism

Idealism has one kind of thing and so escapes the interaction problem between two substances. It still owes an account of order. Separate minds discover the same mathematics, the same primes and the same laws of motion, and an idealist must say what makes them agree without appealing to the mathematics they agree on. Appealing to one universal mind moves the question: the regularities of that mind are then the laws, and laws are structure. Idealism needs mathematics to explain its own order, and mathematics needs no mind to hold. Its mind-stuff is mathematical structure described from inside.

4.5 Mary, Zombies and the Chinese Room

Frank Jackson (1982) describes Mary, who knows all physical facts about color vision through theoretical models but has never seen color, and argues that on first seeing red she learns a fact her physical knowledge lacked. Being a structure and modeling one are two different structures. Knowledge by description models

a pattern in one part of the brain; knowledge by acquaintance instantiates the pattern throughout its sensory systems, and the second is the experience itself. What Mary lacked was that second structure. Acquiring it is a new physical state of her brain, and it needs no new nonphysical property; that is where the identity view and Jackson part ways.

Chalmers (1996) asks whether a physical duplicate could lack experience. If complete physical organization constitutes consciousness, as Section 2.2 argues, the zombie keeps the constitution and lacks what it constitutes, so it is conceivable only the way an unchecked arithmetic error is. A zombie can be imagined; it could not be built.

John Searle (1980) describes an operator manipulating Chinese symbols without understanding them. The systems reply locates understanding, if present, in the organized whole: the room, the rules and the process together. A component's ignorance does not establish the whole system's ignorance. Searle's point stands that producing the right answers need not be enough for understanding. Mathematical monism concerns the complete organization, including the causal relations through which a system learns, acts and relates to its environment, not any program that imitates its output.

5. Objections and Clarifications

5.1 'The Core Premise Is Circular'

Objection: You define 'existence' via 'intelligibility,' which presupposes minds. Is this not anthropocentric?

Response: The objection conflates two senses of 'intelligibility.' *Intelligibility-as-structure:* having determinate properties standing in definite relations, objective and observer-independent. *Intelligibility-as-knowability:* being

graspable by minds, observer-dependent. The claim is not that reality is mathematical because we understand it. It is that anything with determinate properties constitutes structure, and every structure, ruled or random, is one that mathematics describes. Observers require intelligibility-as-structure (Section 2.4), but intelligibility-as-structure does not require observers. The cosmos would remain mathematical even if no consciousness existed.

A deeper form of the charge is that any defense of logic uses logic, so the case begs the question. Hans Albert's trilemma holds that every attempt to ground knowledge ends in a circle, an endless regress or a dogmatic stop (Albert, 1985). Aristotle granted the first point: the principle of non-contradiction cannot be demonstrated, since not everything can be. He showed it by refutation instead: the denier need only say something, and in saying it he has used the principle (Aristotle, 1908, *Metaphysics* IV.4, 1006a5-28). The argument here rests on premises of that kind. It does not assume that reality is mathematical; it starts from what any claim about reality must have, determinate content, and a denial is such a claim.

5.2 'This Is Just a Tautology'

Objection: The claim that intelligible reality must be structured is true by definition and uninformative.

Response: The conclusion is necessary, and it is far from empty. It excludes positions with serious defenders (Section 1), and it reaches them by testing each proposed alternative in turn: substrate, haecceity and non-relational intrinsic properties. A proposed addition must explain what distinguishes it and how it affects anything. If those answers supply further properties and relations, they extend the structure. If they supply none, the addition has no explanatory role. Calling the conclusion necessary does not make it true by definition; Section 2.2

does the work of showing why every definite nature belongs to structure.

Falsifiability, Karl Popper's demand for an observation that could tell against a claim, is a test for contingent empirical hypotheses, claims about the world that could have been false (Popper, 1959). A necessary claim is tested by counterexample instead: one coherent feature that is neither ruled nor random would refute this one. Matheism's empirical side is its measure, a weighting of structures that favors short descriptions (Section 6.2), which predicts short, stable laws and a short rule behind the constants of physics, and a failure there would leave the ontology standing.

5.3 The Gödel Objection

Objection: Does incompleteness, Gödel's result that consistent formal systems capable of arithmetic leave truths unprovable, show that reality or thought must exceed mathematics?

Response: No. Gödel's theorems are themselves mathematical results. The unprovable truths are still mathematical truths, just not provable within that particular formal system. A stronger system of axioms proves them, with new unprovable truths of its own: mathematics is bigger than any one system. For sufficiently expressive, consistent, effectively axiomatized theories, whose axioms a program can list, incompleteness limits what one theory can prove. It does not show that anything escapes mathematics. Undecidability, the existence of statements a system can neither prove nor refute, is inherent to mathematical structure itself, not a limitation that anything escapes by being nonstructural.

5.4 Different Logics

Different logical systems yield different mathematics: classical versus intuitionistic logic (logic without the law of excluded middle), the standard axioms of set theory

(ZFC) versus alternative set theories, paraconsistent logics, in which a contradiction need not imply everything. This does not undermine the thesis. These logical and axiomatic systems are themselves objects of mathematical study. An axiom system can have many models, structures in which its axioms all hold, and different presentations can describe the same structure. Changing axioms therefore does not automatically identify a different physical universe, and our use of classical logic or ZFC does not by itself identify the physical structure we inhabit.

5.5 Is Logic the Basis of Science?

Objection: W. V. O. Quine argued that no statement is immune to revision, the laws of logic included, and Hilary Putnam proposed that quantum mechanics shows the distributive law of classical logic failing for physical propositions (Quine, 1951; Putnam, 1968). If experience can revise logic, logic is no foundation.

Response: Science decides by logic. A theory with its conditions implies a prediction, and an observation that contradicts the prediction refutes the theory by modus tollens, the inference from 'if p then q' and 'not q' to 'not p' (Popper, 1959). Quantum mechanics is stated, solved and tested in classical mathematics, and the lattice of quantum propositions is one more structure inside it (Section 5.4). Quine himself declined to revise classical logic, citing the loss of simplicity and the burden of reasoning within a deviant logic (Quine, 1970). An argument for revising logic must be valid to persuade, so it relies on the logic it would revise. Mathematics, philosophy and science differ in their premises, definitions in the first two and observations added in the third, and they share one logic.

6. What Follows

The identity thesis rests on Sections 2 to 5. The consequences below rest on further premises, stated with each, and they need reasoning alone, with no physics beyond what everyone grants. Each stands on the fact before it, not on that fact's explanation: even if the weighting below is not why our laws are simple, they are simple, and what follows from simplicity still follows.

6.1 Everything Consistent Exists, Fictions Included

Mathematical existence is not a lesser form of existence waiting for an extra act of physical realization. If a complete structure needs no added substrate to constitute it, the identity applies to every mathematical structure. The further premise is that no primitive actuality property selects only some structures as real. A proposed selector has two options. If it describes relations within a larger structure, the structures it does not select keep their mathematical existence. If it does not, it adds an actuality distinction with nothing to explain it.

What a mind can experience is narrower than what exists. Experiencing is processing, and processing takes steps, so a mind can experience only what takes part in the same steps as it does. That is what physical means: every consistent structure exists, and a mind's physics, transitions in mathematics, is what its own steps reach.

Fictions are no exception. A story is a structure: its characters, events and rules stand in definite relations, in an epic, in science fiction or in a manga. Any story one tells is already a piece of logic, and a consistent story exists as that structure. A plot hole is a contradiction, and a contradiction exists only as a description: one can write "2 + 2 = 5" or "a square circle," and no structure satisfies them. The hole is still fixed somewhere. The versions of the story in which some further fact, bolted on, explains the hole are consistent, so they exist: *Deus ex Mathematica*. Which fictions contain a feeling mind is the question asked of brains, and it has

the same answer: those whose structure does what a feeling mind does.

6.2 Why Laws Are Simple

Each bit of a shortest description is random in the sense of Section 2.3: nothing in the description favors 0 over 1, so each bit is a 50/50 choice, and each added bit halves the description's weight. Whether to add another bit at all is a second 50/50, so longer descriptions lose in total as well as one by one. Simple laws dominate. The observed fact stands whatever explains it: our laws fit on a page, and the early universe was smooth. A miracle remains possible on this weighting, as a bolt-on that halves the weight of the world containing it for every bit it adds. Eugene Wigner's "unreasonable effectiveness of mathematics in the natural sciences" is then no puzzle: the world is mathematics, and simple mathematics at that (Wigner, 1960).

6.3 Why Minds Grow in Steps and Time Runs Forward

Minds like ours are complex, so a history that starts simple must grow them, as the Game of Life grows moving patterns from a few cells under four rules. Growth takes causal steps. A cause produces its effect as the next state, and a continuum has no next state: between any two moments lies another. This is an a priori proof in the logician's sense. Bertrand Russell answered that motion is only being at different places at different times (Russell, 1903), but a place for every moment, with nothing that brings about the next, is the emperor's new points. Change without a cause would lie outside logic and reason. On this view continua belong to mathematics as tools of analysis, and whatever happens, happens in steps.

Time is those steps, so it runs forward. An endless history has no last step from which to run back, and a finite reversal must specify the point it turns back from, a

cost the forward history does not pay. A universal code can charge the same length for a finite history and its own reversal, so no penalty is claimed there. Minds run forward too: thinking merges many states into one, and a merged state has many possible pasts, as 2 = x + y has more than one solution.

What happens is a rule applied to a state, giving the next state. The rules are the only things. A structure that only sat there would still exist, as mathematics does, and would do nothing and know nothing, like a dead brain with every connection intact. Stephen Hawking asked what breathes fire into the equations and makes a universe for them to describe (Hawking, 1988). The math rules themselves do. Any further fire would either follow a rule, and so be more of the structure, or follow none, and so be random, which Section 2.3 also counts as structure.

6.4 One Self

On the view taken here, there is one self in every mind, because the bare form of being a subject is the same structure wherever it occurs. Once no separate soul is left, this follows a priori, from what a structure is. Bodies, memories and causal histories distinguish minds. Does each mind also need a primitive owner of experience? If exchanging supposed owners leaves every experience, memory and relation fixed, the exchange makes no structural difference, so Matheism rejects that extra ownership. What remains in each stream is the bare form of being the one who feels, and that form has no feature that differs from one stream to another. A structure is one wherever it occurs, as the number two is one number however many pairs there are, so the self is one while the streams it runs through stay many. This keeps streams of experience distinct; it does not make two brains one object or give them shared memories.

6.5 Agency

An agent can have a determinate decision policy even if it assigns probabilities to several actions. Such a policy is structure, and chance does not remove the agent's character, reasons or causal role. When an agent deliberates, it is doing causal work, not interrupting nature. A free choice that is neither caused nor random is the one option the dichotomy excludes.

7. Conclusion

Reality is mathematical structure, because whatever exists has definite features, every definite feature is ruled or random, and both are mathematical. A further substance either contributes something determinate, which adds structure, or contributes nothing, which explains nothing. Swapping two intrinsic natures while every relation stays fixed changes nothing anyone could notice, think or report, so a nature of that kind could not even be known by the mind it is said to make up. The complete structure is the thing itself.

This is monism. Platonism split the Forms from a world they do nothing in, and Plotinus stacked emanations under a One that has no properties and so is nothing. Matheism keeps one level, and the one it keeps has every property there is.

The hard problem of consciousness assumes its conclusion. Anyone who defines experience so that no structural answer could count has assumed the very split under dispute. The open question for the philosophy of mind is which structures constitute experience.

The claim is necessary, as 2 + 2 = 4 is, and it has teeth: one coherent feature neither ruled nor random would refute it. Its consequences reach far. Every consistent structure exists, fictions included. Simple laws carry the weight, since each added bit halves a description's weight. Minds grow in causal steps from a simple start. Time is those steps, and it runs forward, since an endless history has

no last step to run back from. The bare subject of experience is one self in every mind, with bodies, memories and histories still distinct.

Any search for an ultimate explanation ends at mathematical structure, because structure is the only medium in which an explanation can be given. Asking 'why structure?' is like asking 'why are explanations explanatory?': the question presumes what it questions. The intelligibility of reality and the reality of intelligibility turn out to be the same thing, grasped from different angles.

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